

Literature Status: The Isotropic Mean-Norm Question

Automated mathematics run `fa.banach.001`, lane 11

9 August 2026

Status: literature-implied answer (full answer to the extracted mean-norm question). This is an agent-identified scaling implication from a later theorem, not a new proof by the run.

Original question

G. Chasapis, A. Giannopoulos, and N. Skarmogiannis ask in [1], arXiv:1906.03719, PDF page 3, whether for every volume-one isotropic centrally symmetric convex body $K \subset \mathbb{R}^n$ one has

$$L_K \sqrt{n} M(K) \leq c(\log n)^b$$

for some absolute constants $c, b > 0$. Here

$$M(K) = \int_{S^{n-1}} \|\theta\|_K d\sigma(\theta),$$

and volume-one isotropy means that the uniform random vector X_K on K satisfies

$$\text{Cov}(X_K) = L_K^2 \text{Id}.$$

Supporting theorem

P. Bizeul states the following as Theorem 1.4 of [2], arXiv:2607.29458, PDF page 3, attributing the mean-norm estimate to Bizeul–Klartag together with the newly available dimension-free bound on the third-moment parameter κ_n :

If K_0 is an isotropic convex body in the probabilistic convention $\text{Cov}(X_{K_0}) = \text{Id}$, then

$$M(K_0) \leq C \sqrt{\frac{\log n}{n}},$$

where C is universal.

Identification

Given a body K in the convention of the original paper, set

$$K_0 = L_K^{-1} K.$$

Then $X_{K_0} = L_K^{-1} X_K$ in distribution, so $\text{Cov}(X_{K_0}) = \text{Id}$. For every $a > 0$, gauge homogeneity gives $\|x\|_{aK} = a^{-1} \|x\|_K$. Consequently,

$$M(K_0) = L_K M(K).$$

Applying the supporting theorem to K_0 and multiplying by $\sqrt{n}$ yields

$$L_K \sqrt{n} M(K) \leq C \sqrt{\log n}.$$

Thus the extracted question has an affirmative answer with $b = 1/2$. The supporting paper records sharpness via the cube, so the logarithmic exponent $1/2$ is optimal up to universal constants.

Provenance and scope

The supporting paper does not cite the 2019 question or say that it is answering arXiv:1906.03719. The match is therefore an agent-identified direct reformulation, classified as a *literature-implied answer*. This packet resolves only the explicit mean-norm question on source PDF page 3; it makes no claim about every broader problem discussed in the source paper.

The search used the exact expression $L_K \sqrt{n} M(K)$ and the phrases *isotropic mean norm*, *polylogarithmic*, and $M(K)$, then checked arXiv:2510.20511, 2607.24164, and 2607.29458. The last paper provides the decisive theorem in its final unconditional form.

References

- [1] G. Chasapis, A. Giannopoulos, and N. Skarmogiannis, *Norms of weighted sums of log-concave random vectors*, arXiv:1906.03719 (2019).
- [2] P. Bizeul, *Optimal MM^* bounds for convex bodies*, arXiv:2607.29458 (2026), Theorem 1.4.